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Editorial

Increased Prevalence of Myopia in the United States between 1971 and 1972 and 1999 and 2004—A Reappraisal

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The prevalence of myopia in the United States population was estimated to increase from 25% to 42% over 30 years between the 1971−1972 and 1999−2004 National Health and Nutrition Examination Survey.[1] Nonetheless, there were unexplained anomalies in the data. For example, those aged 18 to 24 years in 1971−1972 (who would be 45−57 years of age in 1999−2004) had myopia prevalence of 28%, but those aged 45 to 54 years in 1999−2004 had myopia prevalence of 45%, suggesting meaningful adult-onset myopia. Here, we reappraise these data and propose a model to account for the 30-year age-dependent changes in prevalence.

We developed a theoretical framework of myopia prevalence by age and time, allowing for both increased prevalence and adult myopia onset and progression between surveys.

Data from the 12- to 17-year age group were not considered as they would be most susceptible to the lack of cycloplegia. By way of illustration, Figure 1A assumes a starting prevalence of 25% among those aged 18 to 24 years in 1971−1972, with no increased prevalence or adult-onset myopia over time. In Figure 1B, an increase of 15% prevalence between surveys is assumed without any adult progression. Figure 1C shows adult-onset myopia of 15% in the absence of change in prevalence over time. Finally, in Figure 1D, both an overall increase of myopia prevalence of 15% and adult onset of 15% are modeled.

An unweighted linear regression was performed to model the influence of sex, change in prevalence across time, and adult-onset myopia using data from Vitale et al.[1] A similar process was performed for a threshold of −2 diopters (D) (their Table 1), with the assumption that any changes in prevalence would be due to adult myopia progression, rather than onset.

There was a significant interaction whereby greater myopia prevalence among women at a younger age disappeared in the older age group. This observation aside, myopia prevalence data are best explained by an increase of 17.1% in overall population prevalence—consistent with Vitale et al[1]—and with 18.9% due to adult-onset myopia (Fig 1E).

For a myopia threshold of −2 D (Fig 1F), an increase of 13.7% in overall population prevalence was accompanied by 11.4% adult myopic progression.

The differences between the National Health and Nutrition Examination Survey data from 1971−1972 to 1999−2004 are best explained by not only an increase in overall population myopia prevalence, but a substantial component of adult-onset myopia and progression. A recent comprehensive review concluded that adult-onset may represent a third or more of all myopia in western populations, but less in East Asia, where onset during childhood is high.[2] The model presented here supports that conclusion with 19% of all individuals developing myopia beyond the age of 18 years. The review also indicates that clinically meaningful myopia progression continues in early adulthood that may average −1 D between the ages of 20 and 30 years. This latter assertion is supported by our recent analysis of 3 studies, including the 1971−1972 and 1999−2004 National Health and Nutrition Examination Survey data,[1,3,4] that all provide evidence of around −1 D myopia progression between the ages of 20 and 50 years.[5]

We have used a simple analysis of summary data here and more sophisticated modeling using source data might be more informative. For example, a nonsignificant trend for greater adult-onset myopia in the 1999−2004 survey in our analysis could be investigated. Furthermore, most adult-onset myopia occurs before 30 years of age; however, our regression assumes a constant incidence rate with age. Nonetheless, our analysis suggests that interpreting the increasing prevalence over 30 years is informed by both substantial adult-onset myopia and adult myopia progression.

Footnotes and Disclosures

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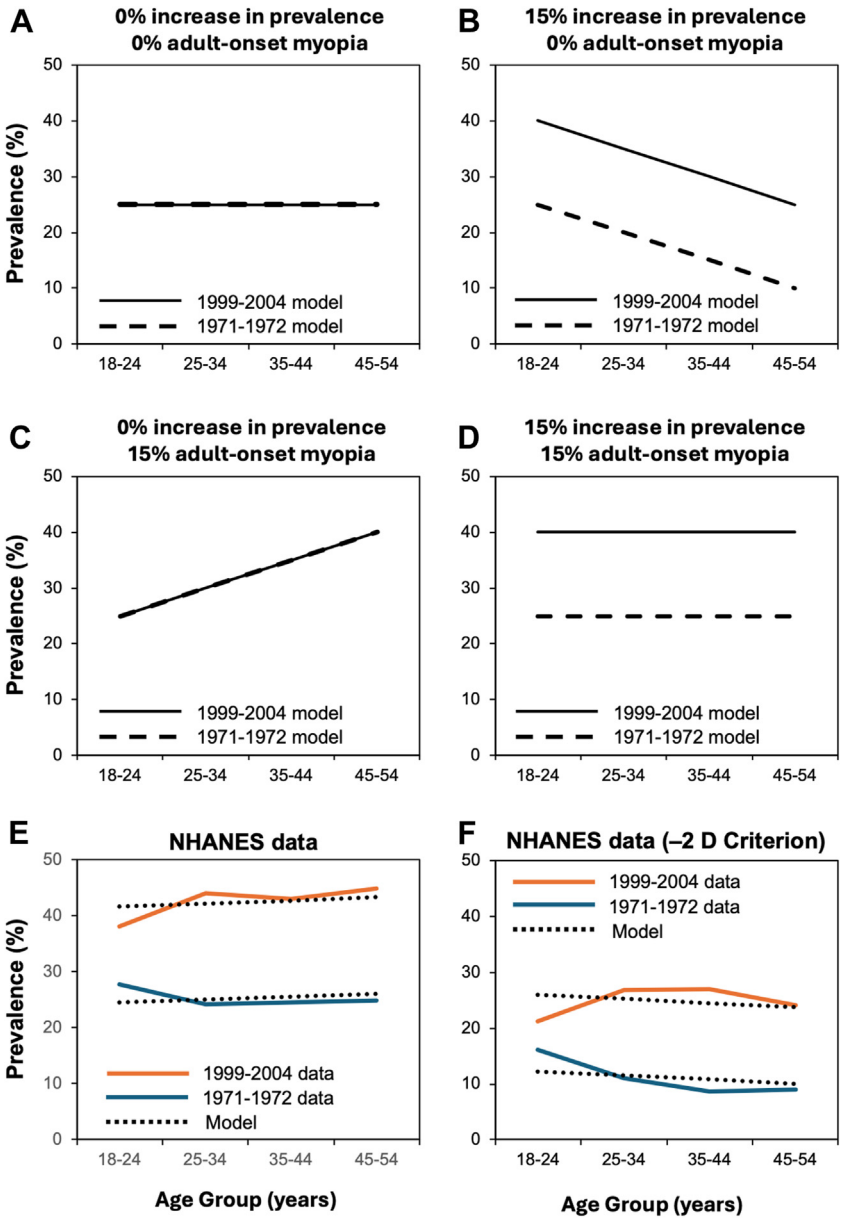


Figure 1. Prevalence of myopia. **A,** No increased prevalence over time or adult-onset myopia. **B,** A 15% increase in prevalence, but no adult-onset myopia. **C,** No increased prevalence over time but 15% adult-onset myopia. **D,** Both a 15% increase in myopia prevalence and 15% adult-onset myopia. **E,** NHANES myopia prevalence data modeled by an increase of 13.0% in prevalence and an increase of 17.1% due to adult-onset myopia. **F,** NHANES myopia prevalence data and modeled by an increase of 8.8% in overall population prevalence for this threshold accompanied by an increase of 13.7% due to adult myopic progression. NHANES = National Health and Nutrition Examination Survey.

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